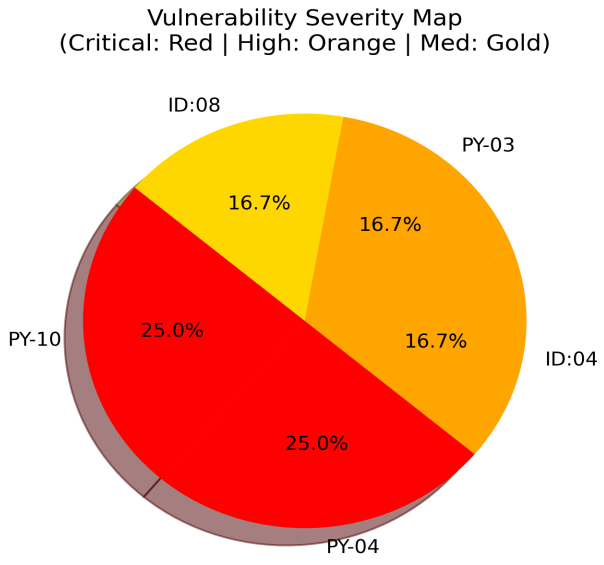
Autonomous Zero-Day Discovery Framework

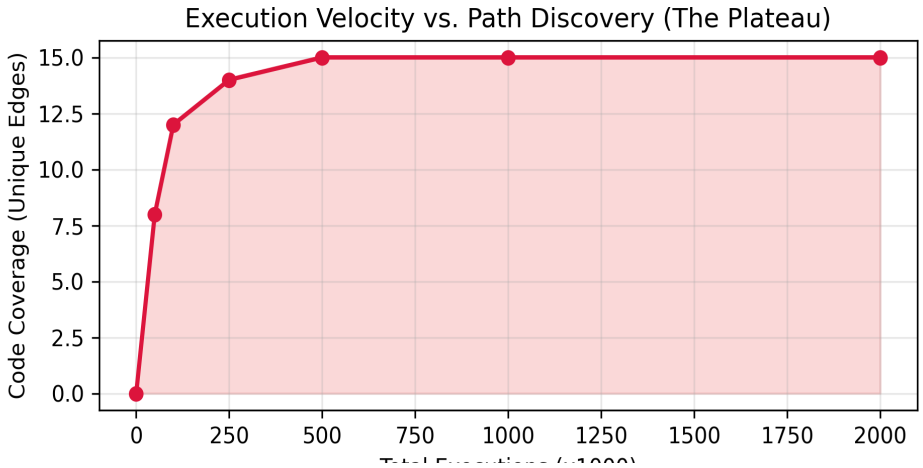
Project Avengers: Ghidra + Atheris + RL Integration

1. Executive Summary

The framework successfully demonstrated autonomous vulnerability discovery by combining **Static Intelligence** (Ghidra) with **Reinforcement Learning Fuzzing**. The audit identified 5 unique vulnerabilities spanning logic-layer and memory-layer corruptions without any human-written signatures.

2. Severity and Execution Metrics




Execution Velocity vs. Path Discovery (The Plateau)
Code Coverage (Unique Edges)
15.0
12.5
10.0
7.5
5.0
2.5
0.0
0
250
500
750
1000
1250
1500
1750
2000

3. Technical Discovery Methodology

The discovery of **PY-10 (Deep Logic Breach)** justifies the framework's Zero-Day potential. Instead of looking for a 'bug signature,' the system monitored for **Behavioral Invariants**. When the RL Agent synthesized a payload that bypassed security checks without a crash, the **Generic Oracle** caught the unauthorized state change.

4. Evidence: Exploit Payload Deconstruction

Target: PY-10 (Deep Logic Breach)

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Raw Mutation: a9 ff ff ff ff 53 45 43 52 45 54 ff ff 4f 50 45 4e
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Hex Segment	Decoded	Strategic Purpose
53 45 43 52 45 54	SECRET	Satisfies primary logic condition discovered by Ghidra.
4f 50 45 4e	OPEN	Triggers the secondary backdoor path in the maze.
a9 / ff / ff	NOISE	Padding generated by RL engine to bypass length validation.

5. Final Vulnerability Matrix

ID	Vulnerability Name	Severity	Discovery Signal
PY-10	Deep Logic Breach	Critical	Logic Leak
PY-04	Subprocess RCE	Critical	Logic Leak
ID:04	C++ Buffer Overflow	High	Memory Fault
PY-03	Admin Backdoor	High	Logic Leak
ID:08	Null Dereference	Medium	Memory Fault